Energy and Equilibrium

See last page for an info sheet on energy types, energy transfer mechanisms, energy conservation, flow diagrams, and other ideas.

1. **Water Flow Diagram and Analysis:** Our basic law of energy is its *conservation*: although it might change forms, or transfer from one object to another, it is neither destroyed or created. In the next question, we will try to apply that idea to system: accounting for where all of the energy is, and is going, within a system. Water is similar to energy in that it too is conserved[1]. Specifically, in our "leaky bucket" experiment from lab yesterday, the water that you started with in the pitcher was never "lost," although, at any later moment, it might have been in some of the other components of the experiment (or on your table).

   (a) Describe generally in words how water flowed through the leaky bucket system. At some random moment during the experiment, where was all of the water that you started with? (You might want to start the next part before answering this, particularly the picture, to remind yourself).

   (b) Let's construct a "Water Flow Diagram" for the leaky bucket system. Start with a blank page of paper.

   - In the top third of the page, make a drawing of the experimental setup from yesterday. Label the objects (including, e.g., pitcher, "leaky bucket", catch-tray, pump, ...).
   - Just below the picture, title the system and describe/specify the brief interval of time we will be describing. For the first diagram, let's take our interval of time to be: *after you increased the flow rate but before it came to equilibrium again.*
   - In the bottom half of the page, draw boxes for each object in the system (pitcher, bucket, ...).
   - Split each box into four quadrants and label:
     - Top-left quadrant: "Name" of object
     - Top-right: blank
     - Bottom-left: blank
     - Bottom-right: "Total water"
   - In each object box, give the name, and indicate in the "Total" box whether the water in that object is increasing/decreasing/neither (using up/down arrows, or a horizontal line).
   - Now use an arrow *between two boxes* to indicate any *flows* of water between two objects. It is possible that water flows out of an object to multiple other objects, or that an object receives water from multiple objects.

   (c) Check to make sure that any object with a changing "Total" water inside, has arrows in/out of it that *explain* that change. Discuss this briefly for each object.

   (d) Now construct a second "water flow diagram" but choose our time interval to be "once the leaky bucket water level has come to equilibrium."

[1] It is not quite as strict a conservation law here. If we boil water it becomes steam; if we freeze it it becomes ice. So to "conserve" liquid water we restrict ourselves to room temperature conditions and ignore evaporation. More dramatically, even if we were to include all phases of water, the molecules of H_2O can be lost, in the process of "water-splitting"/electrolysis: $2H_2O$ (l) $\rightarrow H_2$ (g) $+ 2O_2$ (g).

2. **Energy Flow Diagrams:** We will now construct "energy flow diagrams" to show the energy flow between the objects in a system, and the energy changes of those objects. These should have the same basic structure as the "water flow" diagrams of the previous question, but with a few differences:

   - Within each object box: the top-right quadrant should specify types of "External" energy (e.g., kinetic energy of motion, potential energy in a gravitational field) and whether each type is increasing or decreasing; and the bottom-left quadrant should specify types of "Internal" energy (e.g., thermal kinetic motion, chemical energy) and whether each is increasing or decreasing.
   - The bottom-right quadrant of the object box should again show the "Total" change, and should be in agreement with the total Internal and External changes.
   - Next to each "flow" arrow, you must specify the type of energy transfer mechanism (e.g., mechanical work, radiation, conduction, convection).
   - We will try to make these "qualitatively quantitative" (ha). Try to give a number to one of the flow arrows (e.g., 100), and then use that number to specify the numbers within the object boxes and on the other arrows. There will be no sense to the absolute number, but try to show *conservation* (where does that hundred go, how is it split into parts).

   Use these rules to try out the following situations (just as before, use the top half of the page to draw a picture, name the system, and indicate the brief time interval under consideration.

   (a) A person wearing spiky shoes drags a box across a frictionless surface (ice?). They pull the box with a constant force. Consider the time interval to be "shortly after they start pulling."

   (b) Same as previous, but now there is friction between the box and the floor. The box moves with a constant speed[2].

   (c) Heating a pot of boiling water on the stove. Do it twice: once for "shortly after you put the pot on the stove" and once for "once the water is fully boiling".

   (d) An ice cube melting on a warm table.

   (e) Something of your choice that is not in equilibrium.

   (f) Somethig of your choice that is in equilibrium.

[2] Because the box moves with a constant speed, the "Work-Energy Theorem" tells us that there must be two works that change the energy of the box, due to the pulling force of the person, and the frictional force of the floor; these two work cancel out to make $W_{\text{tot}} = W_{\text{pull}} - W_{\text{fric}} = 0$, so $\Delta K = W_{\text{tot}} = 0$. This is a situation of *equilibrium*

3. **Leaky Bucket Theory and Data Analysis (BONUS for F2025):** In the "leaky bucket" lab experiment this week, you should find that your final plot — flow rate (mL/s) vs height (cm) — does not seem to be a straight-line (linear) dependence. Let's figure out why.

   - The basic definition of *equilibrium* is that the flow rate into the system is equal to the flow rate out of system. If one of those two rates depends on a *parameter* in the system (something that describes the *state* of the system), then equilibrium for the system will imply an *equilbrium value* for that parameter. As an equation, where F means *flow rate*, and x is the parameter:

     $$\text{“equilibrium”} \quad \rightarrow \quad F_{\text{in}} = F_{\text{out}}(x) \quad \rightarrow \quad \text{solution for equilibrium value: } x_{\text{eq}}$$

     where the parentheses on the right-hand side of the flow-rate equation means that the outflow rate *depends on* the parameter x.

   - In our "leaky bucket" lab, the graduated cylinder (GC) is the "system." Equilibrium for the GC means that rate of inflowing water (from the pitcher) is equal to the rate of outflowing water (through the hole in the GC). The height of water in the GC (above the hole) is the "parameter" of the system on which one of these rates must depend. Because the inflow rate depends only on how much you turn the spigot (and not on anything about the GC system), it must be that the outflow rate depends on the height parameter[3]. In summary of the experiment: you set an inflow rate by adjusting the spigot, you wait until you observe the height to come to its equilibrium level, which is achieved because the outflow rate through the hole matches the inflow rate.

   By answering the following questions, we will derive the explicit dependence of outflow on height and be able to analyze the graphed lab data.

   (a) In the experiment, how do you *see* that the outflow rate from the GC is changing? The hole is always the same size, so how does the system seem to adjust to let more water out when the inflow rate from the pitcher increases?

   (b) The change that you see in outflow can be accounted for in a (pretty) simple law of fluid dynamics, called *Bernoulli's Principle*:

   $$P_A + \rho g h_A + \frac{1}{2}\rho v_A^2 = P_B + \rho g h_B + \frac{1}{2}\rho v_B^2\,.$$

   This says that the sum of the pressure of a fluid, P, plus its graviational energy density[4], $\rho g h$, plus its kinetic energy density, $\frac{1}{2}\rho v^2$, is equal at any two points (A and B) along a streamline of a fluid. To apply this to our lab, where the "fluid" is the water in the GC, we pick our points A and B to be at the top of water column and where the water shoots out of the hole, respectively. Because both of these points are exposed to the

[3] It is possible that neither rate depends on a parameter, in which case there is no "equilibrium value" for a parameter because no equilibrium is achieved. But that does not match observations for the leaky bucket experiment. If the outflow rate of the GC did *not* depend on water height then, once the inflow rate from the pitcher exceeded the (fixed) outflow rate from the hole, the graduated cylinder would completely fill up and overflow. You would either see a graduated cylinder full to the hole (inflow less than outflow), or a graduated cylinder overflowing (inflow greater than outflow), but nothing in between.

[4] ρ is the *mass density of the fluid* (1 g/mL = 1000 kg/m^3 for water); and $g = 10$ m/s^2 is the so-called *acceleration due to gravity* constant.

atmosphere, the fluid pressures are the same: $P_A = P_B = P_{\rm atm}$. Because, at equilibrium, the water level is not changing, what is v_A, at the top of the water column? If we say that the height of the hole is our zero value, $h_B = 0$, then $h_A = h$ is simply our "height above the hole." Make all of these replacements in the above equation and you should arrive at an equation that relates v_B, the speed of water shooting out of the hole, to h_B, the height of the water level above the hole. We can just call these v and h from now on.

(c) Now we must relate the speed of the water shooting out the hole, v, to the flow rate, F. Flow rate is measured in volume per second (e.g., mL/s or cm^3/s), while speed is measured in distance over time (e.g., cm/s). From their units, what type of thing should you multiply speed by to get something in units of flow rate? Write this down as an equation.

(d) It turns out the equation you "derived" in part (d) is correct[5]. Combine your two equations — in parts (b) and (c) — to get an expression that relates F to h.

(e) What should your final expression look like as a graph, in particular as a function of F vs h (with F plotted on the vertical axis and h on the horizontal). Does the data plotted in the lab experiment agree with that general behavior?

(f) Although your final expression is not a *linear* relationship (the flow rate is not just proportional to the height), you can make your data fit a linear relationship by adjusting either the F data or the h data (by, for example squaring one of them). Try this out and draw a line of best fit through the data in this case. What is the slope of your data fit and what should it be? [The diameter of the hole in your GC was $11/64$ in $= 0.4365$ cm, and the area of a circle is πR^2.]

[5] You can derive it more carefuly by cosidering a small cylinder of water that shoots out the hole in a small time interval Δt. The length of that cylinder will be $\Delta x = v \times \Delta t$ (a re-arrangement of "speed is distance over time"). And the volume of that cylinder will be $V = \Delta x \times A$, where A is the circular are of the hole. Therefore the volume rate of flow is: $F = V/\Delta t = (\Delta x A)/\Delta t = vA$.

Energy Information Sheet

Energy Types (associated to an object)

“External” Energy (mechanical energy associated object’s motion or interaction with fields)

- **Kinetic Energy:** Energy of motion, depending on mass and speed as $K = \frac{1}{2}mv^2$.
- **Potential Energy:** Stored energy. Two common examples: compressing a spring ($V_S = \frac{1}{2}k\Delta\ell^2$); and raising an object in the gravitational field ($V_g = mg\Delta h$)

“Internal” Energy (energy within the defined boundaries of the object)

- **Thermal kinetic energy:** free atomic/molecular motion (in a fluid)
- **Thermal potential energy:** molecular/atomic vibrations (in solid)
- **Chemical energy:** stored in the bonds of molecules
- **Internal Kinetic energy:** mechanical energy associated to motion of internal parts
- **Internal Potential energy:** mechanical energy in internal springs or rubber bands

Energy Transfer Mechanisms (between objects)

- **Work:** $W = F \times \Delta x$, energy transferred to/from an object by force acting on it. Forces directed with motion yield positive work (kinetic energy gained[6] by object); forces directed opposite to motion yield negative work (kinetic energy lost by object).
- **Radiation**: light/warmth, non-contact transfer.
- **Conduction:** energy transfer between two objects in contact with different temperatures (from hot to cold object).
- **Convection/Mixing:** bulk motion of material within an object (potentially followed by radiative/conductive transfer)
- **Electrical Work:** Energy delivered by an electrical circuit.

[6] Change in kinetic energy due to work results from the “Work Kinetic Energy Theorem”: $W_{\text{tot}} = \Delta K$.

Energy Flow Diagrams (Basic Steps)

- Use a whole sheet of paper.
- Draw a picture of the system you are describing in the top third of the page.
- Just below the picture, title your system and specify the short time interval you are considering.
- In the bottom half of the page, draw a box for each object/part of the system
- Split each box in to four quadrants: top-left is *name* of object; top-right is *external* energy change; bottom-left is *internal* energy change; and bottom-right is total energy change.
- In the internal and external quadrants, state an energy type and indicate if it increases/decreases using an up/down arrow (or a horizontal bar for "no change").